RAJAT KUMAR

Computer Vision and Gen AI Engineer
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PROFESSIONAL SUMMARY

Computer Vision and Gen AI Engineer with 4+ years of experience **designing, optimizing, and deploying deep learning models**. Specializes in 2D/3D pose estimation, generative AI (LLMs/ diffusion /consistency models), text detection, edge-device deployment, and RAG-based multimodal pipelines.

PROFESSIONAL EXPERIENCE

Software Engineer – ML / Consultant — 2022 – Present

Scaledge India Pvt. Ltd. | *Deployed at SONY*

AI/ML Training Analysis Using GenAI & RAG — *Jan 2026 – March 2026*

- Built an **end-to-end pipeline for automated deep learning training analysis** using Generative AI and Retrieval-Augmented Generation (RAG).
- Training logs, validation logs, model config files, and TensorBoard graphs are multi-modal inputs to a GenAI pipeline that produces a structured analysis document.
- **Implemented a dual-index RAG system**: textual content (metrics, config, analysis) stored in a text vector database, and visual data (TensorBoard graphs/images) stored in a separate image vector database.
- Integrated an LLM for the Q&A interface, which uses retrieved RAG context (both text and images) to generate answers about model training behaviour.
- Builted an interactive **Gradio UI based chatbot for the whole pipeline**.

Skills: ***Python, GenAI, RAG, Vector Databases, TensorBoard, LLMs, Multimodal Retrieval, Gradio***

Motion-Based Content Creation — *July 2025 – Dec 2026*

- **Optimized RTMLib (2D pose estimation) codebase** and implemented batch processing, TensorRT (FP32/FP16), and ONNXRuntime inference for RTMW and DWPose models.
- Achieved inference speed improvements of 1.77× (ONNX), 1.13× (TensorRT FP32), and 2.73× (TensorRT FP16).
- Conducted a comprehensive **survey of state-of-the-art 3D motion capture volumetric methods** and implemented selected models on custom data.
- Proposed and implemented **architectural modifications** to Faster VoxelPose, MVGFormer, and TEMPO, **improving qualitative results on custom data without retraining**.

Skills: ***Python, PyTorch, 3D Motion Capture, Faster VoxelPose, MVGFormer, TEMPO, RTMW, DWPose***

Sports AI – 3D Text-to-Motion Generative AI — *April 2025 – June 2025*

- **R&D on 3D text-to-motion generative AI models**; implemented MotionLCM and DiP.
- Removed the VAE component from the architecture and retrained MDM with two alternative prediction modes.
- **Converted the DiP diffusion model to a consistency model** via a **novel architecture change**.
- Improved FID from 12 to 1.65 and R-Precision (Top-1) from 0.23 to 0.40, considered production-quality results.

Skills: ***Python, PyTorch, Diffusion Models, DDPM, DDIM, DiP, MotionLCM***

Text Detection & Recognition in Sports Videos — *March 2024 – March 2025*

- Built an **end-to-end real-time OCR pipeline** for sports broadcast video streams.
- Generated **synthetic training data using Pillow and Generative AI** for the detection task.
- Achieved **93.2% word-level detection accuracy** on full frames and **95.8% on cropped scoreboards** using YOLOv9 (MIT licence).
- **Achieved ~88% recognition accuracy** using the CLOVAAI model.
- Applied **torch.compile to achieve ~5× inference speed improvement with no accuracy degradation**.

Skills: ***Paddle OCR, CLOVAAI, MMOCR, YOLOv4/v9/v10, Pillow, TensorRT, Python***

Sony Mobile Camera Computing Library Development — *July 2022 – March 2024*

- **Optimized AI models for deployment on Sony mobile cameras** using PyTorch, SNPE, QNN, and AIMET (PTQ, QAT, Pruning).
- **Generated and deployed on-device inference libraries via SNPE APIs.**
- Investigated TFLite, NNC, and ONNX frameworks to improve the NN library deployment pipeline.

Skills: ***PyTorch, SNPE, QNN, AIMET, ONNX, NNC, Quantisation, Pruning, C/C++***

Car Licence Plate Detection & Recognition (PoC)

April 2022 – July 2022

- Developed a two-stage pipeline: YOLOv5 for plate detection and OCR for number recognition from images and video.

Skills: ***YOLOv5, OCR, PyTorch, Python***

Junior Research Fellow

2021 – 2022

IIT Mandi

Plant Growth Prediction Using GANs

August 2021 – March 2022

- Generated synthetic future plant growth images from time-series data using the FutureGAN model.
- Correlated Growing Degree Days (GDD) with plant Rate of Growth (ROG) using temperature data.

Skills: ***GANs, FutureGAN, PyTorch, LabelMe, YOLACT, Python***

Weather Forecasting Using LSTM and TCN

January 2021 – August 2021

- Forecast maximum and minimum temperatures using local and global datasets.
- Derived a correlation coefficient to enable prediction of local temperature from global data.

Skills: ***TCN, LSTM, PyTorch, Google Colab, Python***

EDUCATION

B.Tech in Computer Science Engineering2016 – 2020

ABVGIET Shimla | CGPA: 7.5/10

TECHNICAL SKILLS

Languages: Python, C/C++

Frameworks & Libraries: PyTorch, ONNX, TensorRT, ONNXRuntime, OpenCV, Pillow

Model Optimization: Quantization (PTQ/QAT), Pruning, TensorRT FP16/FP32, torch.compile, SNPE, QNN, AIMET

Computer Vision: 2D/3D Pose Estimation, Object Detection (YOLO family), OCR, GANs, Diffusion Models

Tools & Platforms: Google Colab, LabelMe, MMOCR, CLOVAAI, SNPE SDK, NNC

Generative AI: DDPM, DDIM, Consistency Models, MotionLCM, DiP, FutureGAN

LLMs & RAG: Retrieval-Augmented Generation (RAG), Vector Databases, Multimodal Retrieval, Ollama, Deepseek local setup